

Fluid Dynamics and The Plateau-Rayleigh Instability

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I Introduction

The summer project was spent studying the basic equations and concepts of fluid dynamics in a more mathematically sophisticated manner. After studying the fundamentals of fluid dynamics, the tools were applied to studying the Plateau-Rayleigh Instability. The main references used were *The Physics of Fluids and Plasmas—An Introduction for Astrophysicists* by Professor Arnab Rai Choudhuri, as well as various resources on the internet, including MIT Web Lecture notes on the Plateau-Rayleigh Instability.

2 The Governing Equations of Fluid Dynamics

Like all areas of physics, the subject of Fluid Dynamics is described completely by a set of governing equations. These include the continuity equation and the Navier-Stokes equation, which are respectively given below:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0 \quad (1)$$

$$\rho \frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = \mathbf{F} - \frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{v} \quad (2)$$

Here, ρ is the density of the fluid, $\mathbf{v}$ is the velocity of the fluid element, p is the pressure, and ν is the coefficient of the kinematic viscosity, which is equal to μ/ρ where μ is the coefficient of viscosity.

The Continuity Equation describes the property of fluids that mass does not disappear on a local scale. For unit volume, the change in mass is equal to the sum of the change in density with time, which is the first term of the equation, and the movement of mass into or out of the unit volume, which is the second term.

The Navier-Stokes Equation is the equivalent of Newton's equation $\mathbf{F} = m\mathbf{a}$ for classical mechanics in that it describes the effect of various forces acting on a unit volume of the fluid. In many applications, such as the derivation of the Plateau-Rayleigh Instability, the effect of viscosity is ignored, giving what is called the Euler Equation:

$$\rho \frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = \mathbf{F} - \frac{1}{\rho} \nabla p \quad (3)$$

Another common assumption made in problems is that of the incompressibility of the fluid, i.e. $\frac{\partial \rho}{\partial t} = 0$.

3 Preliminary Tools and Concepts

3.1 Young-Laplace Equation

The Young-Laplace equation, which describes the effect of surface tension, is the following.

$$\Delta p = -\gamma \nabla \cdot \hat{\mathbf{n}}$$

where p is the pressure across the surface, γ is the coefficient of surface tension, and $\hat{\mathbf{n}}$ is the unit normal vector to the surface.

It is also written as

$$\Delta p = \gamma \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \quad (4)$$

where R_1 and R_2 are the principal radii of curvature. This form is more convenient in many cases.

3.2 Principal Curvatures

The Young-Laplace Equation refers to two principal radii of curvature. These are the radii corresponding to the principal curvatures of the surface (at the given point).

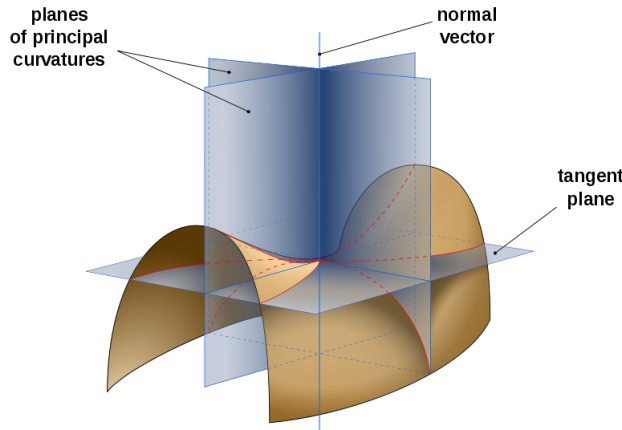


Figure 1: A diagram to help illustrate the principle radii of curvature.

For any point on the surface, an infinite number of normal planes can be drawn which contain the unit vector normal to the surface at that point. Each of these planes intersect with the surface along a curve. For example, the curve created by the intersection of

the curved surface of a cylinder and the plane perpendicular to the cylinder axis (which contains the normal vector at that point) is a circle.

The principal curvatures are the minimum and maximum curvatures which can be obtained by varying the normal plane at the point.

3.3 Modified Bessel Functions

Modified Bessel functions are function which satisfy differential equations of the form

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - (x^2 + \alpha^2)y = 0 \quad (5)$$

Here $\alpha \in \mathbb{C}$ is called the order of the Modified Bessel function.

Solutions to the Modified Bessel differential equation are of two types, I_α and K_α . I_α grows exponentially from 0 at $x = 0$ and K_α decays exponentially from ∞ at $x = 0$.

For the analysis of the Plateau-Rayleigh Instability, only I_α will be needed. An expression for I_α for integer α is

$$I_\alpha(z) = \frac{1}{\pi} \int_0^\pi e^{z \cos \theta} \cos(\alpha \theta) d\theta$$

An important relation is

$$I'_0(z) = I_1(z) \quad (6)$$

4 The Plateau-Rayleigh Instability

The Plateau-Rayleigh Instability describes the phenomenon where a falling jet or cylinder of fluid (such as water) at one point ceases to be a jet and breaks into multiple droplets of smaller total surface area. The short explanation for this is that the surface energy is minimized by this change, but the Plateau-Rayleigh Instability can be analyzed more quantitatively and in greater detail.

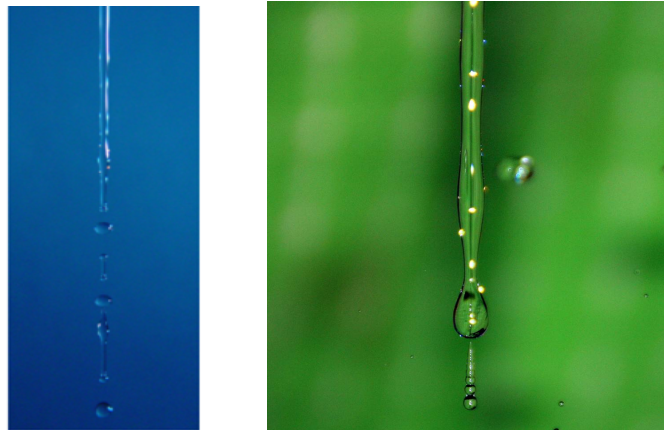


Figure 2: Jets of water breaking into droplets.

The phenomenon is primarily one of surface tension. For this reason gravity and viscous forces are not significant and hence are ignored.

4.1 The Unperturbed State

As the effects of viscosity and gravity are being ignored, the unperturbed cylinder of water will not change shape as it falls (with gravity, we expect the radius of the cylinder to decrease with the increasing velocity of the water). Hence, the unperturbed state is a infinite cylinder of radius R_0 .

4.2 An Intuitive Explanation

The cylindrical jet of water will be subject to perturbations of different wave number components. One component is illustrated in Figure 3, where its crests and troughs are seen.

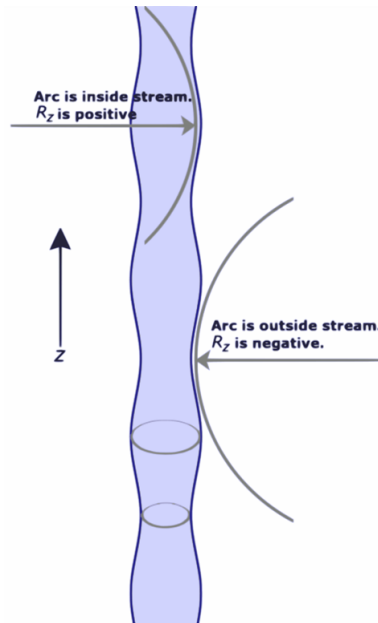


Figure 3: A simplified diagram showing the opposing effects of surface tension.

From equation (4), we see that surface tension depends on the two radii of curvature; in the case of the cylinder, R_1 is the radius of the cylinder at that point, while R_2 is the radius of curvature of the waves along the z -axis.

Thus, $1/R_1$ is higher at troughs than at crests, creating a pressure which pushes the fluid from troughs to crests. However, R_2 shows the opposite relation: R_2 is positive at crests and negative at troughs, which opposes the contribution of R_1 in (4).

Hence the two contributions oppose each other. The phenomenon depends on which of the two contributions dominates over the other. The components that grow with time and cause the break-up of the jet are those for which the radius of curvature of the cylinder is greater than that of the wave.

4.3 Quantitative Analysis

The fluid is assumed to be non-viscous and incompressible, i.e. $\gamma = 0$ and $\partial\rho = 0$.

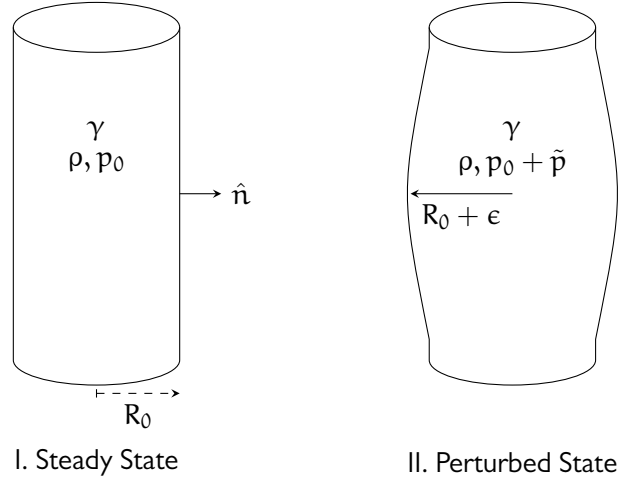


Figure 4: The steady state and perturbed state.

Let the steady state cylinder of fluid have radius R_0 . Then the perturbed radius can be represented as

$$\tilde{R} = R_0 + \epsilon e^{\omega t + i k z} \quad (7)$$

Here $\epsilon \ll R_0$, k is the wave number, and ω is the angular frequency (which may be complex). Due to symmetry considerations $\tilde{R}$ cannot be dependent on the angular position and hence there is no θ dependence in the expression.

We set the radial and vertical components of the perturbation velocity as u_r and u_z , and use the Euler equation to get conditions on them.

In cylindrical coordinates, the r and z components of the Euler equation are (setting all angular terms to zero)

$$\begin{aligned} r : \quad & \rho \left(\frac{\partial u_r}{\partial t} + u_r \frac{\partial u_r}{\partial r} + u_z \frac{\partial u_r}{\partial z} \right) = - \frac{\partial p}{\partial r} \\ z : \quad & \rho \left(\frac{\partial u_z}{\partial t} + u_r \frac{\partial u_z}{\partial r} + u_z \frac{\partial u_z}{\partial z} \right) = - \frac{\partial p}{\partial z} \end{aligned}$$

We now linearize these two equations by approximating all terms involving more than one perturbed variable (such as u_z) to zero to get

$$r : \quad \frac{\partial u_r}{\partial t} = - \frac{1}{\rho} \frac{\partial p}{\partial r} \quad (8)$$

$$z : \quad \frac{\partial u_z}{\partial t} = - \frac{1}{\rho} \frac{\partial p}{\partial z} \quad (9)$$

We get one more equation by using the continuity equation in cylindrical coordinates and linearizing it,

$$\frac{\partial u_r}{\partial r} + \frac{\partial u_z}{\partial z} + \frac{u_r}{r} = 0 \quad (10)$$

4.3.1 The Ansatz

At this point we make an ansatz by anticipating that the radial and vertical components of the perturbation velocity as well as the pressure must have the same form as the perturbation itself. This is reasonable on physical ground as the pressure is the quantity governing the perturbation, including its radial and vertical velocity components. Thus

$$u_r = R(r)e^{\omega t + ikz} \quad (11)$$

$$u_z = Z(r)e^{\omega t + ikz} \quad (12)$$

$$\tilde{p} = P(r)e^{\omega t + ikz} \quad (13)$$

Substituting these into (8)–(10), we get

$$\frac{\partial P}{\partial r} = -\rho\omega R \quad (14)$$

$$ikP = \rho\omega Z \quad (15)$$

$$\frac{\partial R}{\partial r} + \frac{R}{r} + ikZ = 0 \quad (16)$$

By differentiating (15) and (16) with respect to r and hence eliminating P and Z ,

$$r^2 \frac{\partial^2 R}{\partial r^2} + r \frac{\partial R}{\partial r} - (1 + k^2 r^2) R = 0 \quad (17)$$

This is of the form of (5), i.e. a Modified Bessel equation of order 1 in the variable kr . Thus the solution is

$$R(r) = CI_1(kr) \quad (18)$$

for an as-yet undetermined constant C . Here we have not included a component of the Modified Bessel Function of the second type as we need $R(0) = 0$ in order to have a smooth radial velocity profile at $r = 0$.

4.3.2 The Boundary Conditions

The constant C can be determined by applying appropriate boundary conditions. The first is the kinematic relation, which equates the rate of change of the radius $\tilde{R}$ with the radial component of the perturbation velocity, u_r , on the free surface, i.e. at $r = R_0$.

$$\begin{aligned} \frac{\partial \tilde{R}}{\partial t} = u_r &\implies \epsilon\omega e^{\omega t + ikz} = R(R_0)e^{\omega t + ikz} \\ &\implies \epsilon\omega = CI_1(kR_0) \end{aligned}$$

$$\Rightarrow \boxed{C = \frac{\epsilon \omega}{I_1(kR_0)}} \quad (19)$$

We get one more condition by equating the internal and external pressures on the free surface. From the Young-Laplace equation (4),

$$p_0 + \tilde{p} = \gamma \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \quad (20)$$

Here $R_1 = \tilde{R} = R_0 + \epsilon e^{\omega t + ikz}$. As $\epsilon \ll R_0$,

$$\frac{1}{R_1} = \frac{1}{R_0 + \epsilon e^{\omega t + ikz}} \approx \frac{1}{R_0} \left(1 - \frac{\epsilon}{R_0} e^{\omega t + ikz} \right) \quad (21)$$

$-1/R_2$ is the curvature of the surface at R_0 , i.e.

$$\frac{1}{R_2} = -\frac{\partial^2 \tilde{R}}{\partial z^2} = \epsilon k^2 e^{\omega t + ikz} \quad (22)$$

p_0 is the equilibrium pressure when there are no perturbations. Hence $R_1 = R_0$ and $R_2 = \infty$, giving $p_0 = \gamma/R_0$.

From (20), (21), (22), we get

$$\begin{aligned} \frac{\gamma}{R_0} + \tilde{p} &= \gamma \left(\frac{1}{R_0} - \frac{\epsilon}{R_0^2} e^{\omega t + ikz} + \epsilon k^2 e^{\omega t + ikz} \right) \\ \Rightarrow \tilde{p} &= -\frac{\gamma \epsilon}{R_0^2} \left(1 - k^2 R_0^2 \right) e^{\omega t + ikz} \end{aligned} \quad (23)$$

From the relation between $P(r)$ and $R(r)$ in (14) and the value of C in (19), and that $I_1(x) = I_0'(x)$,

$$\begin{aligned} P(r) &= -\frac{\epsilon \omega^2 \rho}{I_1(kR_0)} \int_0^r I_1(kx) dx \\ &= -\frac{\epsilon \omega^2 \rho}{k I_1(kR_0)} I_0(kr) \\ \Rightarrow \tilde{p} &= -\frac{\epsilon \omega^2 \rho}{k I_1(kR_0)} I_0(kr) e^{\omega t + ikz} \end{aligned} \quad (24)$$

Now equating this value of $\tilde{p}$ with the one obtained in (23), we obtain the dispersion relation:

$$\omega^2 = \frac{\gamma}{\rho R_0^3} k R_0 \frac{I_1(kR_0)}{I_0(kR_0)} \left(1 - k^2 R_0^2 \right) \quad (25)$$

Because of the $(1 - k^2 R_0^2)$ factor on the RHS, we see that ω can only be real (and hence cause unbounded growth of the wave component, leading to rupture) if

$$kR_0 < 1$$

The maximum value of ω will be the component that grows fastest and causes the rupture. This maximum value is attained for

$$kR_0 \approx 0.697$$

5 Standing Capillary Waves

The preceding analysis assumed an infinite cylinder with no obstructions. However, it is observed that in the presence of an obstacle, standing capillary waves are formed, as can be seen in Figure 5.

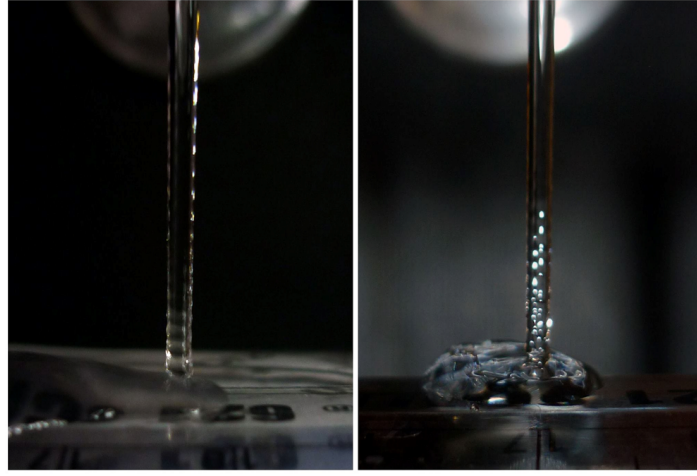


Figure 5: Standing Capillary Waves.

Given the velocity of the stream, the wave number and hence frequency of the wave can be determined by using the dispersion relation (25) and using $\omega^2 = U^2/k^2$ where U is the (known) stream velocity.

$$U^2 = \frac{\omega^2}{k^2} = \frac{\gamma}{\rho k R_0^2} \frac{I_1(kR_0)}{I_0(kR_0)} (1 - k^2 R_0^2)$$

6 Acknowledgments

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